Travis Aerenson

Taerenson.github.io * 302-690-7030 * Taerenso@uwyo.edu

Education

University of Washington, Seattle, WA
Dec 2023
PhD, Atmospheric Sciences
Dissertation Title: Cloud Changes in Climate Models: Response to Solar and CO2 Forcing and the Relationship between Model Bias and Feedbacks

University of Washington, Seattle, WA
May 2021
Master of Science, Atmospheric Sciences
Thesis Title: When Will MISR Detect Rising High Clouds?

Colorado College, Colorado Springs, CO
May 2019
Bachelor of Arts, Physics

Research Experience

Associate Research Scientist **U Wyoming Department of Atmospheric Science, Laramie WY** *Oct 2025-Present*

Postdoctoral Scholar **U Wyoming Department of Atmospheric Science, Laramie WY** *Jan 2024-Oct 2025*

Research Assistant, **U Washington Department of Atmospheric Sciences, Seattle WA**
- Advised by Dr. Roger Marchand *Aug 2019-Dec 2023*

Research Intern, **National Center for Atmospheric Research, Boulder CO** *Jun -Aug 2018*

Research Intern, **National Center for Atmospheric Research, Boulder CO** *Jun -Aug 2017*

Teaching Experience

Guest Lecturer, **UWyo Department of Atmospheric Science, Laramie WY** *Oct 2024-Present*
- ATSC 2100: Global Warming: The Science of Humankind's Energy Consumption Impacting Climate
- ATSC 5040: Global Physical Climatology

Teaching Assistant, **UW Department of Atmospheric Sciences, Seattle WA** *Mar 2021-Jun 2021*
- ATM S 100: Climate Justice & Energy Solutions

Quantitative Reasoning Center Math and Physics tutor, **Colorado College, Colorado Springs CO** *Mar 2017-May 2019*

Publications

In prep:
Aerenson, T., McCoy, D., Nugent, J., Brown, H., Schlaff, G.: Convective vs large scale precipitation: how distinct processes in models combine to set precipitation intensity. *Journal of Climate*

Werapitiya, G., **Aerenson, T.**, McCoy, D., Brient, F., Elsaesser, G., Song, C., & Zelinka, M.: How the hydrological cycle shapes global cloud feedback. *PNAS*

In review:
Aerenson, T., McCoy, D., Keys, P., & Caulton, D.: Precipitation over the Contiguous United States is coming from farther away than in the past. *Geophysical Research Letters*

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Aerenson, T., Barahona, D., & McCoy, D.: Using a testbed model to identify sources of bias in spaceborne cloud droplet number concentration retrieval and propose new alternatives. *JGR: Atmospheres*

Published:

Nugent, J. M., Brown, H., Kirby, A., McCoy, D. T., Allen, G., **Aerenson, T.,** Burrows, S. M., Caulton, D., Fan, J., Feng, Y., Gettelman, A., Griswold, J., Jones, D. B., Leung, L. R., Mahfouz, N., Mikkelsen, A., Mülmenstädt, J., Qian, Y., Shan, Y., … Zhang, K. (2026). Overview of the Nephele Perturbed Parameter Ensemble for Aerosol-Cloud Interactions in E3SMv3. *Journal of Advances in Modeling Earth Systems*, *18*(2), e2025MS004989. https://doi.org/10.1029/2025MS004989

Aerenson, T., McCoy, D., Elsaesser, G., Wu, J., Nugent, J. M., Brown, H., Mikkelsen, A., & Zelinka, M. D. (2025). Global models predict clouds at the wrong time of day: Does it matter for radiation and climate? *Science Advances*, *11*(50), 3236. https://doi.org/10.1126/SCIADV.ADY3236

Werapitiya, G., McCoy, D. T., Elsaesser, G. S., Gettelman, A., Eidhammer, T., **Aerenson, T.**, Song, C., & Wu, J. (2025). Extratropical Cloud Feedback Constrained by Cloud Sources and Sinks in Cyclones. *Journal of Climate*, *1*(aop). https://doi.org/10.1175/JCLI-D-24-0607.1

Song, C., McCoy, D., Molod, A., **Aerenson, T.,** & Barahona, D. (2025). Signatures of aerosol-cloud interactions in GiOcean: a coupled global reanalysis with two-moment cloud microphysics. *Atmospheric Chemistry and Physics*, *25*(21), 15567–15592. https://doi.org/10.5194/ACP-25-15567-2025

Aerenson, T., McCoy, D., Elsaesser, G., & Wu, J. (2025). Drivers of Model Spread in Snowpack Changes across the American Mountain West. *Journal of Climate*, *38*(22), 6605–6619. https://doi.org/10.1175/JCLI-D-25-0103.1

Aerenson, T. & Marchand, R. (2025). How Do Differences in the Simulation of Present-Day Clouds Affect Cloud Feedbacks? *Journal of Geophysical Research: Atmospheres*, *130*(12), e2025JD044020. https://doi.org/10.1029/2025JD044020

Aerenson, T., Marchand, R., & Zhou, C. (2024). Cloud Responses to Abrupt Solar and CO2 Forcing: 2. Adjustment to Forcing in Coupled Models. *Journal of Geophysical Research: Atmospheres*, *129*(12), e2023JD040297. https://doi.org/10.1029/2023JD040297

Aerenson, T. & Marchand, R. (2024). Cloud Responses to Abrupt Solar and CO2 Forcing: 1. Temperature Mediated Cloud Feedbacks. *Journal of Geophysical Research: Atmospheres*, *129*(12), e2023JD040296. https://doi.org/10.1029/2023JD040296

Poletti, A. N., W Frierson, D. M., **Aerenson, T.,** Nikumbh, A., Carroll, R., Henshaw, W., & Scheff, J. (2024). Atmosphere and ocean energy transport in extreme warming scenarios. *PLOS Climate*, *3*(2), e0000343. https://doi.org/10.1371/JOURNAL.PCLM.0000343

Aerenson, T., Marchand, R., Chepfer, H. & Medeiros, B. (2022). When Will MISR Detect Rising High Clouds? *Journal of Geophysical Research: Atmospheres, 127(2)*, e2021JD035865. https://doi.org/10.1029/2021JD035865

Aerenson, T., Tebaldi, C., Sanderson, B. & Lamarque, J.F. (2018). Changes in a suite of indicators of extreme temperature and precipitation under 1.5 and 2 degrees warming. *Environmental Research Letters* https://doi.org/10.1088/1748-9326/aaafd6

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Grants and Proposals

Funding Agency: Department of Energy Established Program to Stimulate Competitive Research (DOE-EPSCoR)
Role: Co-I
Status: *Selected*
Title: Creating the framework for the next generation Energy Exascale Earth System Model (E3SM) at PROCEED (Perturbed physics ensemble Regression Optimization Center for ESM Evaluation and Development)
Amount Requested: $3M

Funding Agency: NASA Research Opportunities in Space and Earth Science (ROSES) (2024)
Role: PI
Status: *Not selected*
Title: Snow Energetics and SWE in the SnowEx Campaigns and Models (SESSCaM)
Amount Requested: $730,813

Service

Service Positions:

Postdoc Representative, **UWyo Department of Atmospheric Science, Laramie, WY.**
Oct 2024 – Present

Strategic Planning Committee Member, **UWyo Department of Atmospheric Science, Laramie, WY.**
May – Sep 2025

Peer-to-peer Mentoring Coordinator, **UW Department of Atmospheric Sciences, Seattle WA.**
Jul 2022 – Dec 2023

Diversity and Inclusion Group Coordinator, **UW Department of Atmospheric Sciences, Seattle WA.**
Jul 2021 – Mar 2023

Campus Sustainability Fund Intersectional Sustainability Board, **University of Washington, Seattle WA.**
Aug – Oct 2020

Service Activities:

Panelist for University of Washington College of the Environment Pathways to Postdoc Panel Discussion *Nov 2025*

Panelist for University of Wyoming College of Engineering and Physical Sciences Graduate Program Panel *Sep 2024*

Refereed Manuscripts for the Following Journals: *Journal of Geophysical Research: Atmospheres, JAMES, PLOS ONE, Atmospheric Chemistry and Physics (ACP), Earth and Space Science, Nature: Communications Earth and Environment, Geophysical Research Letters (GRL), Nature Communications, Journal of Geophysical Research: Machine Learning and Computation.*

Reviewed Proposals for the Following Programs: *NASA Postdoctoral Program (NPP): Earth Science, NASA Postdoctoral Program (NPP): Technology Development*

Scientific Presentations

Aerenson, T. 2025: "Snowpack and atmospheric moisture transport: how the water cycle is changing in a warming world." University of Wyoming Department of Atmospheric Science Seminar, Laramie, LY. *Seminar*

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Aerenson, T., D. McCoy, G. Elsaesser, J. Wu, J. Nugent, H. Brown, M. Zelinka, S. Burrows 2025: "Are we simulating clouds at the wrong time of day?" Gordon Research Conference on Radiation and Climate, Lewiston, ME. *Poster*

Aerenson, T., D. McCoy, G. Elsaesser, J. Wu, J. Nugent, H. Brown, M. Zelinka, S. Burrows 2025: "Are we simulating clouds at the wrong time of day?" Gordon Research Seminar on Radiation and Climate, Lewiston, ME. *Oral*

Aerenson, T., R. Marchand, T. Ackerman 2024: "MISR Observed Trends in Cloud Top Height and Constraints on High Cloud Altitude Feedbacks" AGU Fall Meeting: Terra: 25 Years of the Earth Observing System Flagship Observatory, Washington, DC. *Poster*

Aerenson, T., D. McCoy, G. Elsaesser 2024: "Causes of Model Spread in Predictions of Hydroclimate in the Mountain West" AGU Fall Meeting: Hydroclimate and Extremes in the Western United States in a Changing Climate, Washington, DC. *Poster*

Aerenson, T., D. McCoy 2024: "How Much Does the Cloud Diurnal Cycle Impact SWCRE?" Micro2Macro Origins of Climate Change Uncertainty Workshop, Laramie, WY. *Poster*

Aerenson, T., D. McCoy, G. Elsaesser 2024: "How Might we Improve Predictions of Regional Hydroclimate" Oxford Workshop on Model Uncertainty, Oxford, UK. *Oral*

Aerenson, T., D. McCoy, G. Elsaesser 2024: "Can We Do Better at Predicting Regional Hydroclimate" CESM Workshop, Boulder, CO. *Oral*

Aerenson, T. 2023: "The Relationship Between Simulated Present-Day Cloud Attributes and Cloud Feedbacks" University of Washington Department of Atmospheric Sciences Colloquium, Seattle, WA. *PhD Defense*

Aerenson, T., R. Marchand, C. Zhou 2023: "Cloud Adjustments to Solar and CO2 Forcing in Coupled Models" CFMIP Meeting, Paris, FR. *Poster*

Aerenson, T., R. Marchand 2023: "The Contribution of Mean-State Bias to Cloud Feedbacks in Climate Models" CFMIP Meeting, Paris, FR. *Poster*

Aerenson, T., R. Marchand 2023: "The Contribution of Mean-State Bias to Cloud Feedbacks in Climate Models" University of Wyoming Department of Atmospheric Science Seminar, Laramie, WY. *Invited Seminar*

Aerenson, T., R. Marchand 2023: "Using ISCCP and MISR Satellite Simulators to Understand Cloud Feedbacks" NASA GSFC CPC Seminar, Greenbelt, MD. *Virtual Invited Seminar*

Aerenson, T., R. Marchand, C. Zhou 2022: "Cloud Response to Abrupt Changes in Solar Forcing and CO_2 Concentrations" AGU Fall Meeting: Advances in Solar Radiation Modification Research, Chicago, IL. *Poster*

Aerenson, T., R. Marchand, C. Zhou 2022: "Cloud Response to Abrupt Changes in Solar Forcing and CO_2 Concentration" University of Washington Department of Atmospheric Sciences Seminar on Atmospheric Physics and Chemistry, Seattle, WA. *seminar*

Aerenson, T., R. Marchand 2022: "Cloud Response to Abrupt Changes in Solar Forcing and CO_2 Concentrations" CFMIP Meeting: Cloud Processes and Radiative Feedbacks, Seattle, WA. *Oral*

Aerenson, T. 2021: "Cloud Rapid Adjustments and Feedbacks to Abrupt Changes in Solar and CO_2 Induced Forcings" AGU Fall Meeting: Advances in Climate Engineering Research. *Virtual Poster*

Aerenson, T., R. Marchand 2021: "Cloud Rapid Adjustments and Feedbacks to Abrupt Changes in Solar and CO_2 Induced Forcing" CFMIP Meeting. *Virtual Poster*

Aerenson, T. 2021: "When Will MISR Detect Rising High Clouds?" University of Washington Department of Atmospheric Sciences Physics and Chemistry Seminar. *Virtual Seminar*

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Aerenson, T., R. Marchand 2021: "Time of Emergence: When Will We See High Clouds Get Higher?" AMS Annual Meeting. *Remote oral presentation and discussion session*

Aerenson, T., R. Marchand, 2020: "Time of Emergence: When do Climate Models Predict Rising Cloud-Top-Height (CTH) Should be Detected by MISR?" CFMIP Meeting on Clouds, Precipitation, and Climate Sensitivity. *Remote submitted slide and discussion session*

Aerenson, T., R. Marchand, 2020: "When will we see high clouds get higher?" MISR Science Team Meeting, Pasadena, CA. *Oral*

Aerenson, T., 2019: "Climate Models and Climate Change Reversibility" Colorado College Physics Department Senior Seminar Series, Colorado Springs, CO. *Seminar*

Aerenson, T., C. Tebaldi, B. Sanderson, J.F. Lamarque, 2017: "Climate Extremes in Low Warming Scenarios" NCAR CGD Integrated Assessment Modelling Weekly Meeting, Boulder, CO. *Oral*

Scholarships and Awards

Certificate of Distinguished Service 2022: University of Washington Department of Atmospheric Sciences

Top Scholar Award 2019: University of Washington Department of Atmospheric Sciences